ABUL HASNAT ABDULLAH

Robotics & Autonomy · B.Sc. Mechanical Engineering, BUET

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PROFILE

Third-year Mechanical Engineering student at BUET (CGPA 3.95/4.00) working on state estimation and autonomous navigation for ground and aerial robots. I lead the software and autonomy sub-team of Team Interplanetar, BUET's Mars rover team, and have started an undergraduate thesis on uncertainty-aware multi-agent, multi-sensor fusion. Looking for a robotics internship in autonomy, estimation, manipulation or simulation.

Interests: state estimation and sensor fusion · autonomous navigation · UAV and UGV autonomy · robotic manipulation · digital twins and simulation

EDUCATION

Bangladesh University of Engineering and Technology (BUET), Dhaka Jul 2022 – expected 2028

B.Sc. in Mechanical Engineering · CGPA **3.95 / 4.00** (Level 3, Term 1)

HSC, Notre Dame College (GPA 5.00/5.00) · SSC, RAJUK Uttara Model College (GPA 5.00/5.00)

RESEARCH

Undergraduate thesis: uncertainty-aware multi-agent, multi-sensor fusion 2026 – present

Supervisor: Dr. Kazi Arafat Rahman, Professor, Department of Mechanical Engineering, BUET

- Modelling how sensor uncertainty propagates through state estimation, and how to fuse measurements consistently when several robots share them.
- Background from VENTRA: fusing LiDAR, optical flow, range and barometer data in PX4's EKF2, with uncertainty set per measurement.

EXPERIENCE

Team Interplanetar, BUET Mars Rover Team Apr 2025 – present

Sub-Team Lead, Software & Autonomy (Jul 2026 –) · Technical Lead, ERC Remote (Jan – Jul 2026) · Member (Apr 2025 – Jan 2026)

- Lead the sub-team that builds the rover's ROS 2 autonomy stack: localization, state estimation, path planning and waypoint navigation with Nav2.
- Set up 3D LiDAR-inertial mapping with a Livox Mid-360 and FAST-LIO, and visual-inertial odometry with a ZED stereo camera; maps stream live to a handheld ground controller.
- Wrote a Nav2 costmap plugin in C++ that turns FastDEM elevation layers (slope, step, roughness) into traversability costs.
- Took the six-joint arm from CAD to URDF, configured MoveIt 2 for inverse kinematics, and built the operators' arm control GUI.
- Built a Gazebo digital twin of the rover, and brought up swerve (CAN bus) and differential drive bases under ros2_control.
- Configured the long-range radio link and network between rover and base station; integrated object detection for mission tasks.
- Competitions: ERC Remote, Australian Rover Challenge, Anatolian Rover Challenge.

PROJECTS

VENTRA: autonomous UAV for GPS-denied indoor exploration and tunnel inspection 2026

ME 366 term project, team of 4 · PX4, ROS 2, Nav2, SLAM Toolbox, Gazebo

- Fed the 2D LiDAR SLAM pose back into PX4's EKF2 as external vision at 10 Hz, ending the position resets of flow-only flight; held 1.15 m altitude with a 7 cm standard deviation over a 283 s GPS-free flight.
- Developed a SLAM-free shaft inspection mode that takes the bore centre from every scan as an absolute position fix: 24.8 m autonomous descent and return in PX4 SITL, 1–5 cm centre error across five shaft shapes.
- Added frontier exploration with Nav2, a braking-limited avoidance filter, 3D mapping from stacked scans, and an open-source PX4 flight-log analyzer.

Other projects: warehouse AGV simulation (Nav2 with kinodynamic RRT* replanning in Gazebo) · closed-loop drone simulation (PX4 and ArduPilot SITL with ROS 2) · robot arm visualizer (URDF, joint states, workspace) · voice-controlled TurtleBot3 (speech to ROS 2 motion)

TECHNICAL SKILLS

Robotics: ROS 2, Nav2, MoveIt 2, ros2_control, PX4, Gazebo, RViz2, SLAM Toolbox, FAST-LIO

Estimation & perception: Kalman filtering (PX4 EKF2), LiDAR-inertial and visual-inertial odometry, sensor fusion, OpenCV, PyTorch

Programming: C++, Python, C, Bash, CMake, Git, Docker, Linux

Hardware: Pixhawk, Raspberry Pi, microcontrollers, CAN bus, 2D/3D LiDAR, stereo cameras

Mechanical: SolidWorks (CSWP, CSWA), AutoCAD, Fusion 360, ANSYS, FDM 3D printing

AWARDS & CERTIFICATIONS

- Champion, Robo Soccer Challenge, Intra-BUET Robo Challenge 2024 (team lead, RC soccer robot)
- SOLIDWORKS Certified Professional in Mechanical Design (CSWP), Dassault Systèmes, 2026
- SOLIDWORKS Certified Associate in Mechanical Design (CSWA), Dassault Systèmes, 2024
- Self Driving and ROS 2, three-course series (82.5 h): localization, odometry and control, planning and navigation, 2025–26

LEADERSHIP & ACTIVITIES

Co-founder and CTO, Ongikar (non-profit, 2021 – present) · Design leads at BUET Automobile Club and BUET Robotics Society · Watercolour artist (Aquarelle Verse)